Physic

1 -> (3)

2 -> (4)

3 -> (3)

4 -> (3)

5 -> (4)

6 -> The ratio of speed of light in vacume ~~ang~~ and speed of light in that medium is known as reflective index.
speed of light in absolut midium = 1.5×10^8 m/s
" " " " vacume = 3×10^8 m/s

~~Vaa~~ speed of light = $\dfrac{\text{reflective index in vacume}}{\text{reflective index in absolut transparent medium.}}$

$$= \frac{3 \times \cancel{10^8}}{1.5 \times \cancel{10^8}} = 2 \text{ m/s.}$$

7. The fuse
chances of
appliance
pass throu
circut
~~ua~~ fuse
a large
circuit
circut

8. normal
correcte

$\frac{1}{f} =$

$\Rightarrow \frac{1}{f} =$

$\Rightarrow \frac{1}{f} =$

$\Rightarrow f =$

$\Rightarrow P$

$= P$

7. The fuse used in the series to reduce the chances of burning of any electrical appliance when large amount of electricity pass through the it, it brake the circuit when ~~jule~~ joule heating takes place fuse melt. If it replaced by the a large rating then fuse don't melt and circuit continued and due to continued circuit application catch fire.

8. normal vision = u = -25 cm
corrected " = v = -75 cm

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{f} = \frac{1}{-75} - \frac{1}{-25}$$

$$\Rightarrow \frac{1}{f} = \frac{-1 + 3}{-75} = \frac{2}{75}$$

$$\Rightarrow f = \frac{75}{2} = 37.5 \text{ cm} = 0.375 \text{ m}$$

$$\Rightarrow P = \frac{1}{f} = \frac{1}{0.375} = \frac{1000}{375}$$

$$\therefore P = \frac{8}{3} = 2.67 \text{ D}$$

deffect is hypermetropia.
Power = 2.67 D
focal length = 37.5 cm

9. $R = \rho \frac{l}{A}$

given, $l = 2l$
$A = \frac{A}{2}$
ρ = constant

(i) $R = \rho \frac{2l}{A/2}$

$= \rho \frac{2l \times 2}{A}$

$= \rho \frac{4l}{A}$ or $= 4 \rho \frac{l}{A}$

$R_n = 4R$

(ii) resistivity remains same because it does not depends on length and cross-section.

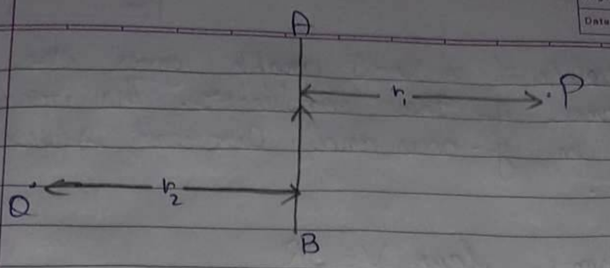
10. (i) P puts you into the plane of paper, and Q takes you out of it. The strength of the magnetic field is larger at the point which is closer to ~~magnet~~ Source, which is Q.

Q

(ii) a) By
shows
b) The dir
center
perpend

11.
(i) In
R

In



(ii) a) By using Campous ~~niddled~~, whose niddle shows the direction.

b) The direction of magnetic field at the center of a current-carrying loop is perpendicular to the plane of the loop.

11.

(i) In case of $10\,\Omega$ R_1

$R_1 = 10\,\Omega$

$V = 3$

then, $I = \frac{V}{R} = \frac{3}{10}$

$I = 0.3$

In case of $15\,\Omega$ R_2

$R_2 = 15\,\Omega$

$V = 3$

then, $I = \frac{V}{R} = \frac{\cancel{3}^{1}}{\cancel{15}_{5}} = \cancel{0.2}$

$= 0.2\,\Omega$

(ii) The solope of a V-I graph gives the resistance of the conductor. The voltage showes in y-axis and Current showes in x-axis.

(iii) By Ohm's law $V = IR$

$\Rightarrow R = \frac{V}{I}$ = slope of V-I graph

Therefor, the solope of the V-I graph represents the effective resistance.

Now in series combination of the resistance all the resistance get summed up.

$R_{eq} = R_1 + R_2 + R_3 + \ldots + R_n$

Therefor, the series combination will have the maximum resistance then the other resistors.

Hence, the line will maximulm slope linee C will give the series combination of the other two resistors.

12. given are, $u = -25\text{ cm}$
$f = -15\text{ cm}$
$h_1 = 4\text{ cm}$
$v = ?$
$h_o = ?$

(i) using mirror formula $= \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\Rightarrow \frac{1}{v} + \frac{1}{-25} = \frac{1}{-15}$

$\Rightarrow \frac{1}{v} = \frac{1}{-15} + \frac{1}{25} \Rightarrow \frac{1}{v} = \frac{-5+3}{75} = \frac{-2}{75}$

$\Rightarrow \frac{1}{v} = \frac{-2}{75} \Rightarrow v = \frac{-75}{2} = -37.5$

$v = -37.5$

(ii) $\frac{h_1}{h_o} = \frac{-v}{u} \Rightarrow \frac{4}{h_o} = \frac{37.5}{-25\text{cm}} \Rightarrow \frac{-100}{37.5}$

$h_1 = \frac{4 \times 37.5}{25} =$

$h_1 = -\frac{150}{25} = -6\text{ cm}$

(iii) 

Real, Inverted